EECS 484 Database Management Systems
Homework 3
Due on Tuesday, March 17th @ 11:55 PM Eastern time

Please read the following instructions before starting the homework:

This homework must be completed individually and can be submitted on Gradescope. Use entry code **D6KXG8** to self-enroll if you don't have access to the Gradescope course page.

No late days for homework! *If you miss the due date, you get 0 points. If your PDF gets modified after the due date, you get 0 points. No exceptions on this.*

Honor Code

By submitting this homework, you are agreeing to abide by the Honor Code:
I have neither given nor received unauthorized aid on this assignment, nor have I concealed any violations of the Honor Code.

Question 1: RA (20 points)

Consider the following schema for this question. Primary keys [INTEGER] are <u>underlined</u> and attributes in **bold** are foreign keys.

Schema:

Passengers(<u>PassengerID</u>, FirstName)
Tickets(<u>TicketID</u>, **PassengerID**, **FlightID**, SeatNumber)
Planes(<u>PlaneID</u>, Airline, Capacity)
Flights(<u>FlightID</u>, **PlaneID**, Destination, DepartureTime)

Write **relational algebra** expressions for the following queries.

1. Find the PassengerID of all passengers who have taken at least one flight with Capacity smaller than 50 or greater than 300. (5 points)

2. Find the FlightID of all flights that have a passenger whose first name is 'Allen', except those flights whose Airline is 'Delta'. (5 points)

3. Find the first name of all passengers who have taken at least one flight with DepartureTime before (<) 22. (5 points)

4. Find the destinations that have been traveled to by all planes. (5 points)

Question 2: Normalization (16 points)

For this part, consider the following functional dependencies.

F = { school → major;
id → name, GPA, major, school;
GPA → major;
name, major → id }

1. List all of the candidate keys for this relation using only the functional dependencies provided (4 points)

2. Is the relation 3NF? Justify your answer (4 points)

3. Is the relation BCNF? Justify your answer (4 points)

4. Consider the following snapshot of the table:

id	**name**	**GPA**	**major**	**school**
17	Alice	3.5	CS	Engineering
32	Bob	3.5	CS	Engineering
43	Christine	3.0	Aerospace	Engineering
67	David	3.8	English	LSA
78	Eve	4.0	Math	LSA
90	Alice	3.8	Astronomy	LSA

Which of the functional dependencies are violated in the snapshot of the table provided? (4 points)

Question 3: Normalization II (13 points)

Relation R has six columns (A, B, C, D, E, F) **and its minimal keys are AD and BD**. The functional dependencies are as follows

- $AC \longrightarrow F$
- $A \longrightarrow B$
- $DF \longrightarrow CE$
- $BD \longrightarrow ACE$

Answer the following questions

1. Which of the following does the functional dependency $A \longrightarrow B$ satisfy? (2 points)
   a. Neither BCNF nor 3NF
   b. BCNF only
   c. 3NF only
   d. Both BCNF and 3NF

2. Which of the following does the functional dependency $BD \longrightarrow A$ satisfy? (2 points)
   a. Neither BCNF nor 3NF
   b. BCNF only
   c. 3NF only
   d. Both BCNF and 3NF

3. Write the first given functional dependency that violates BCNF. If none, write None. If there is, perform one BCNF decomposition on that FD. Give the resulting two tables R1 and R2, and their functional dependency sets F1 and F2 (from the 4 given FDs), respectively. (5 points)

4. Does the above decomposition satisfy the dependency-preserving property? (2 points)
   a. Yes
   b. No

5. Does the above decomposition satisfy the lossless-join property? (2 points)
   a. Yes
   b. No

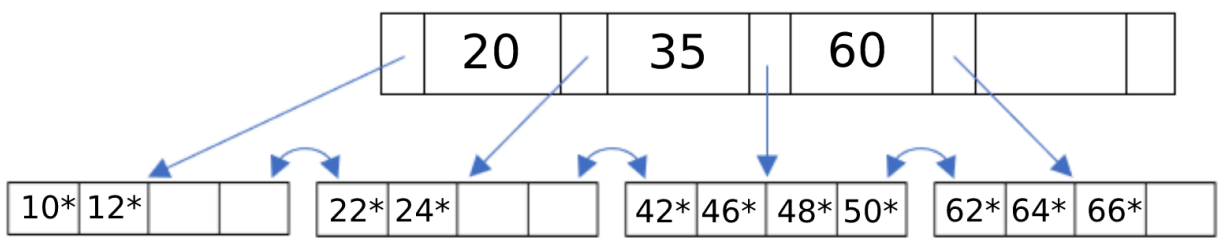
The file *HW3_templates.docx* includes several B+ tree structures that may be helpful for drawing the diagram. Whether you use them or not won't affect your HW score as long as your diagrams are clear and legible.

Question 4: B+ Tree Operations (22 points)

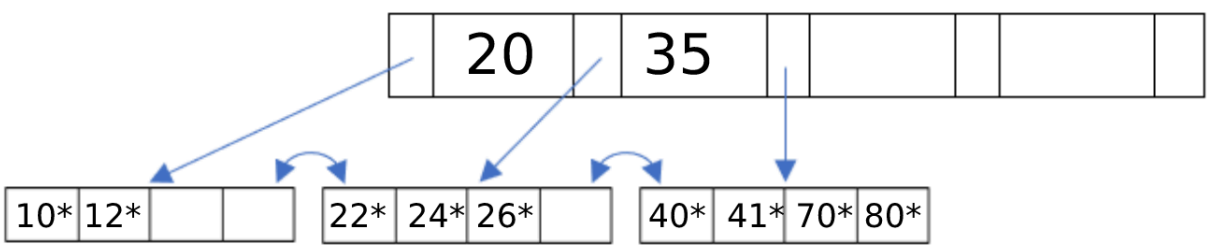
Assume these rules for B+ tree operations in this question:

- The order of both inner nodes and leaf nodes is 2.
- The left pointer points to values that are strictly less than (<) the key value.
- During redistribution, shift elements to/from the right node over the left when both are possible redistribution partners.
- During the splitting of a node, the right node will have one more value than the left node.
- For insertions, favor redistribution over splitting.
- For deletions, favor redistribution over merging.
- Only redistribute one entry at a time.

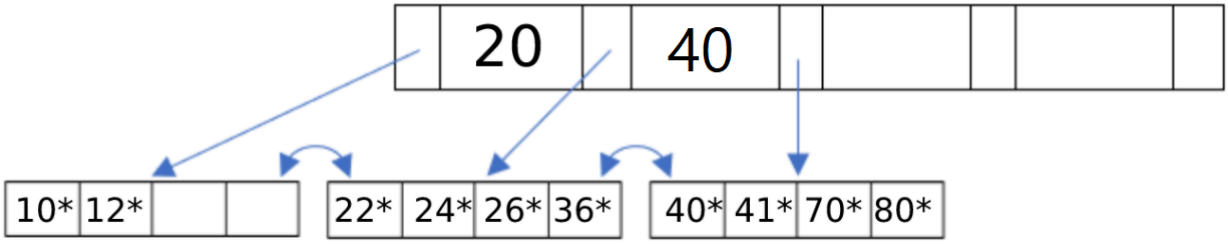
1. Given the tree below, draw the tree after inserting 58* (3 points)



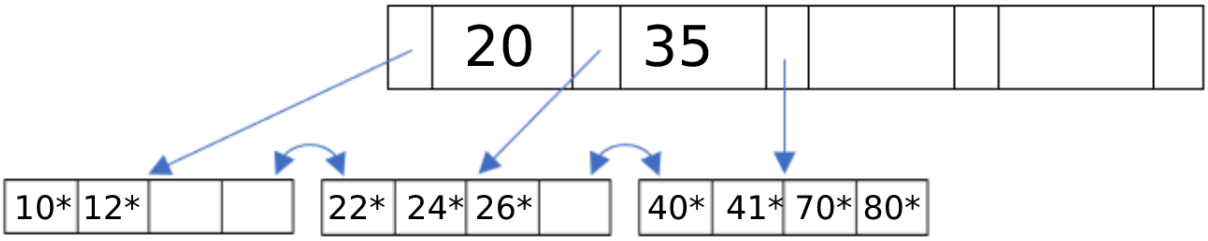
2. Given the tree below, draw the tree after inserting 25*, 30* (3 points)



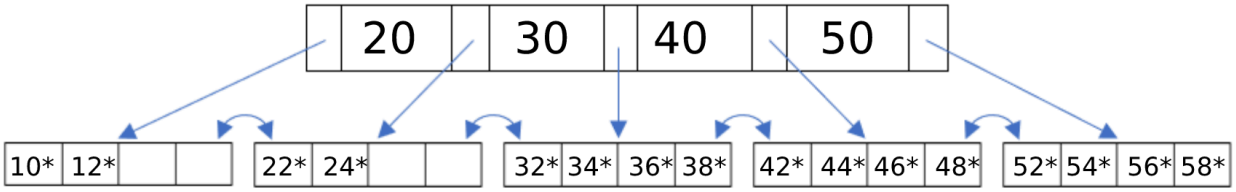
3. Given the tree below, draw the tree after inserting 60* (3 points)



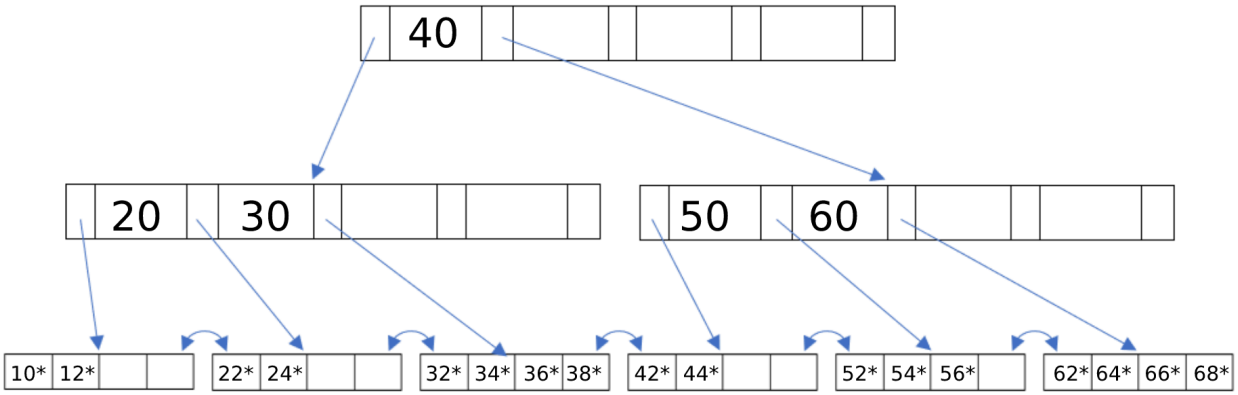
4. Given the tree below, draw the tree after deleting 12* (3 points)



5. Given the tree below, draw the tree after inserting 47* (5 points)



6. Given the tree below, draw the tree after deleting 44*, 52* (5 points)



Question 5: Linear Hashing (24 points)

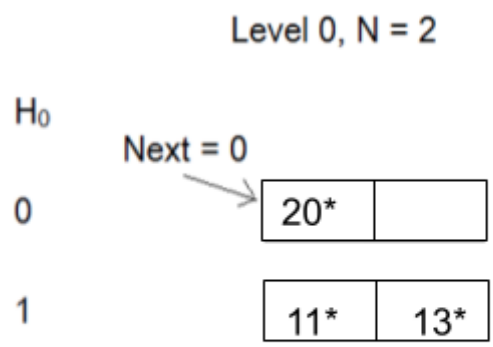
For Question 5, assume the split policy used is splitting when inserting into an overflow page.

The hash function that will be used:

$$h_i(value) = value \bmod (2^i N)$$

Assume that we use the same algorithm discussed in the lecture.

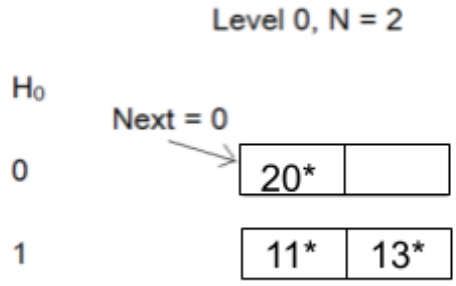
1. We initialize the Linear Hashing index as follows:



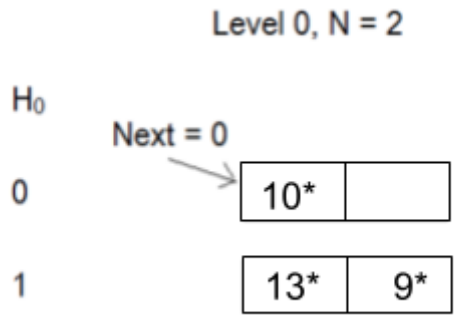
What is the minimum number of insertions needed such that the index structure has 4 buckets and is at Level = 1? Give an example insertion sequence for your answer. (4 points)

2. For each subquestion, draw what the given index structure looks like after the listed insertion. Remember to show the "next" pointer and level post-insertion. (20 points)

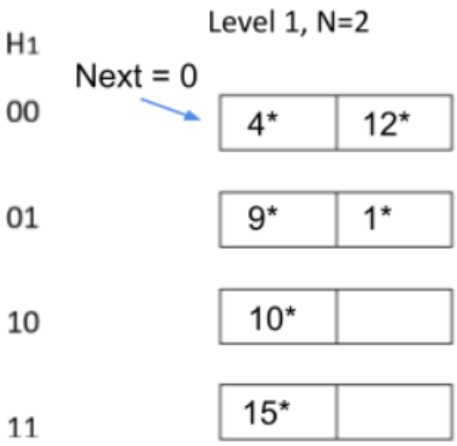
2.1. Insert 22 = $(10110)_2$ (4 points)



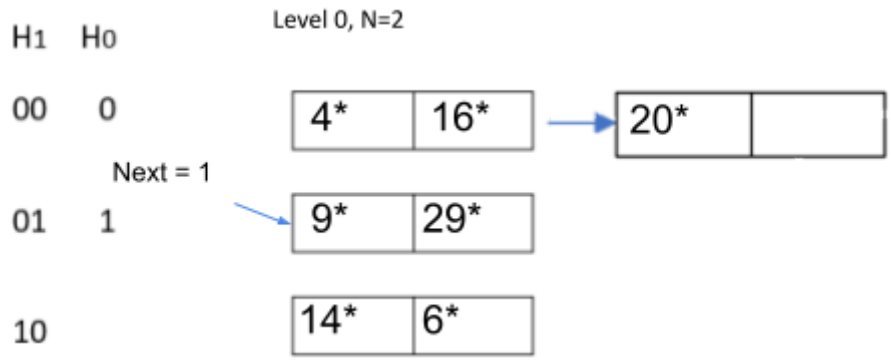
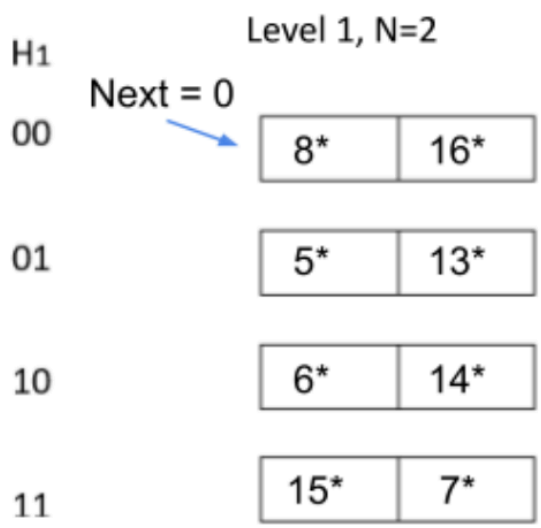
2.2. Insert 23 = $(10111)_2$ (4 points)



2.3. Insert 19 = $(10011)_2$ (4 points)



2.4. Insert 12 = $(01100)_2$ (4 points)



2.5. Insert 20 = $(10100)_2$ (4 points)